

Nanotechnology



**Genetic vision
TEAM**

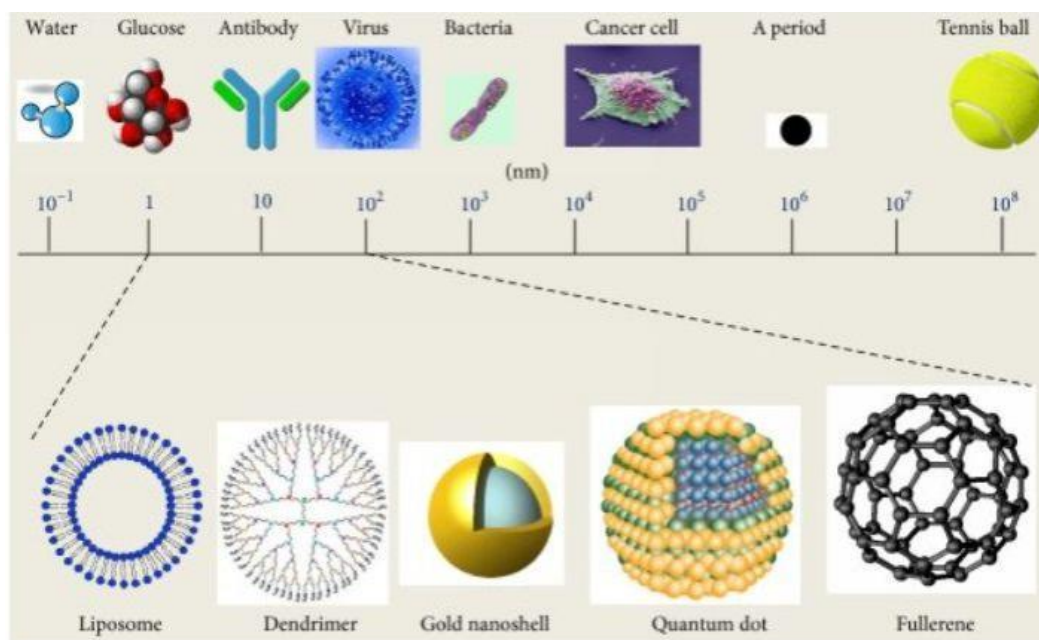
Nanotechnology

Definition : the study of the controlling of matter on an atomic and molecular scale .

Generally nanotechnology deals with structures between 1-100 nanometer in at least one dimension.

Nanotechnology involves developing or modifying material or devices within that size.

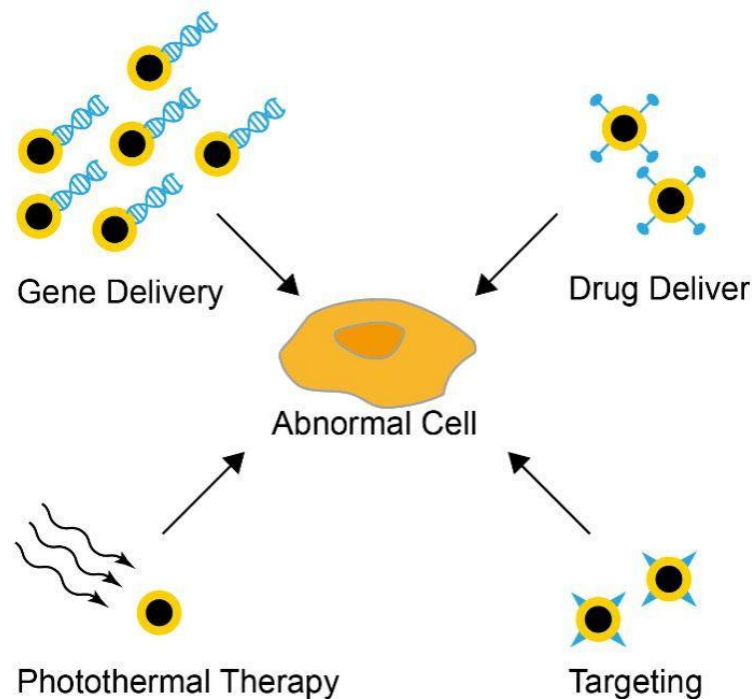
Nano molecules compared to other molecules



Liposome : a spherical vesicle composed of phospholipids and compatible with a lipid bilayer structure use as Delivery method.

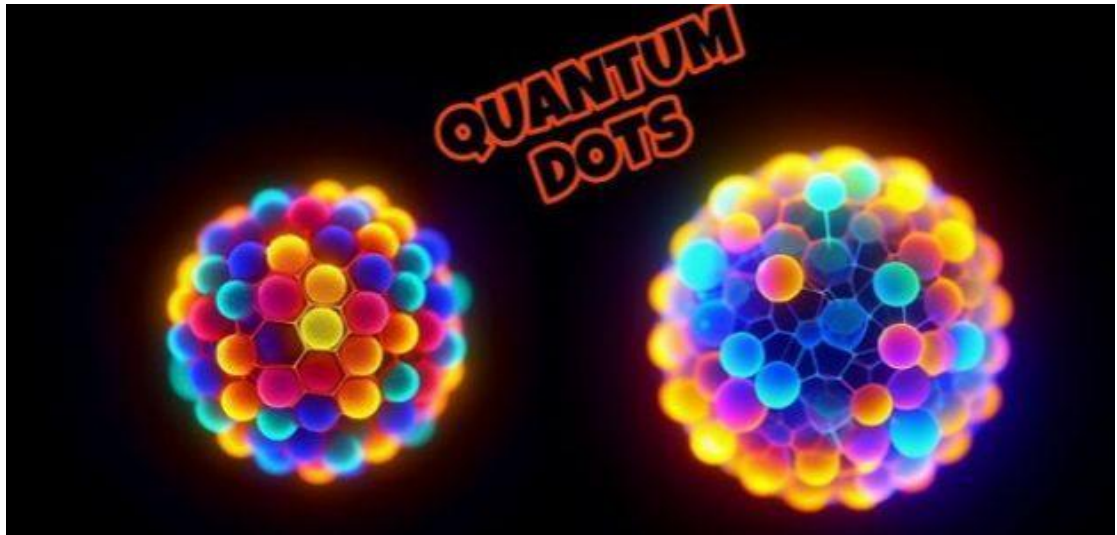
Dendrimer : synthetic polymer with a structure of repeatedly branching chains, typically forming spherical macromolecules use as delivery method.

Gold nanoshell : composed of a silica core coated by a thin gold metallic shell use for optical imaging and bioimaging studies, which help in cancer detection, treatment and medical biosensing.

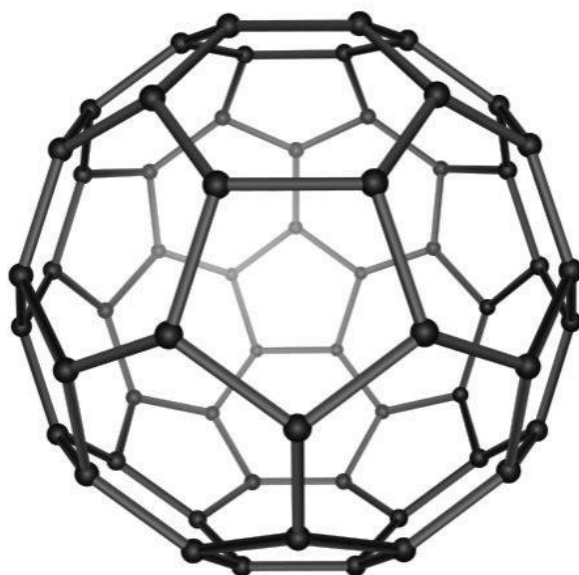


Quantum dots : are semiconductor particles a few nanometres in size with optical and electronic properties that differ from those of larger particles .

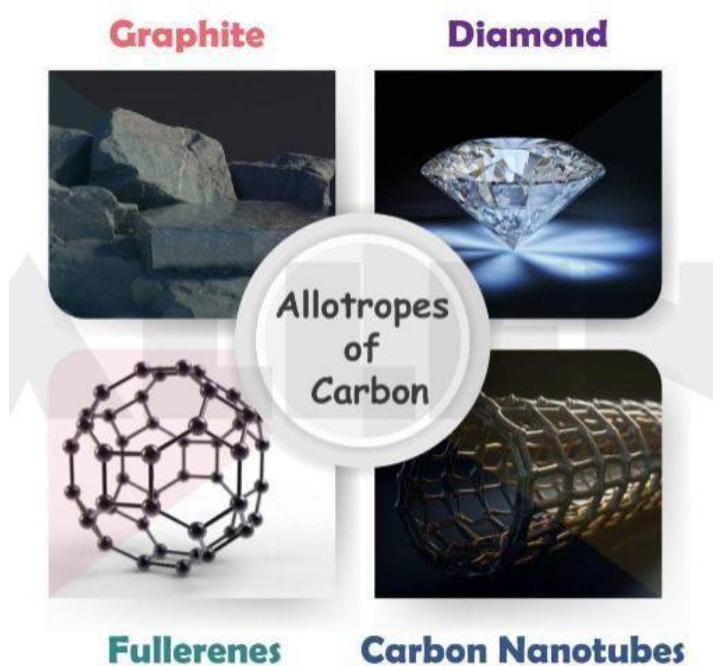
In simple words when the Quantum dots exposed to UV light this causes to Exciting electrons from state to another state, when the electron back to original state the energy that was gotten from the exciting will release and this will lead to emission of light (chemiluminescence).

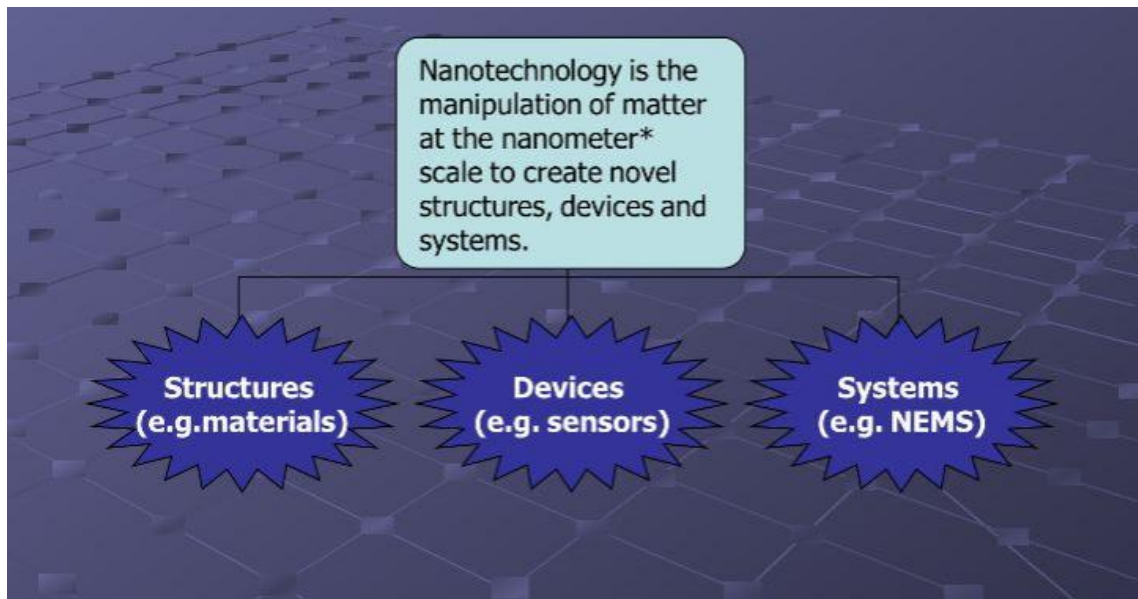


Fullerene : an allotrope of carbon whose molecules consist of carbon atoms connected by single and double bonds so as to form a closed or partially closed mesh, with fused rings of five to six atoms.



Allotropes : one or more forms of a chemical element that can exist in the same physical state.

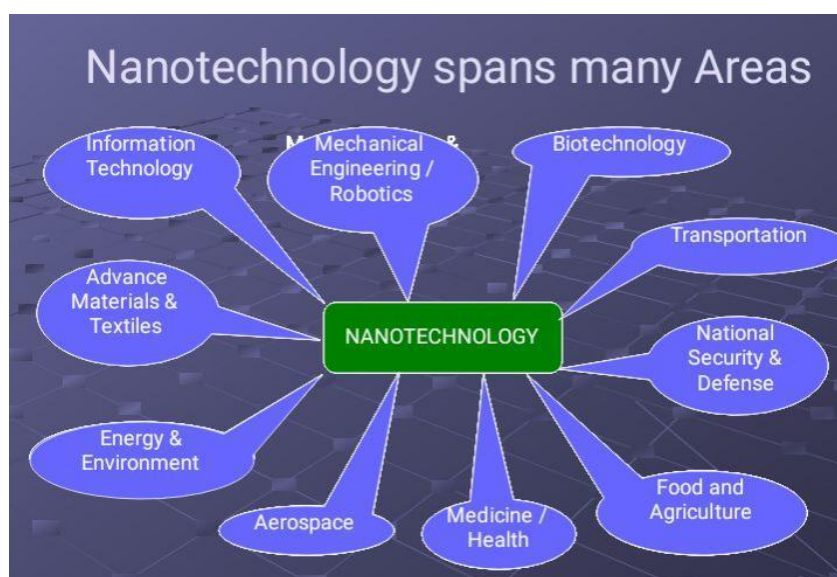




Nanotechnology is already making today's products

- Lighter
- Stronger
- Faster
- Smaller
- More durable

The human hair is about 100,000 micrometer wide



Applications of nanotechnology

1. Diagnosis: the unique optical and physical properties of nanoparticles to manufacture a highly specific and accurate sensors that use to detect the specific analyte.
2. Drug delivery : by using a synthetic molecules in nanometer scale as a transporter for drugs to target cells for example the chemotherapy is used as a treatment for cancer, rather than the strength of chemotherapy to destroy the cancer cells the chemotherapy cannot distinguish between normal cells and cancer cells and also chemotherapy kill the normal cells by using I'm in Berlin for example liposome molecule or synthetic delivery molecule which in Nano-scale to deliver the chemotherapy which it provide a highly specificity to deliver the drug for cancer cells only.
3. Tissue engineering : can be used to make a scaffolds for tissue engineering, promoting cell growth and differentiation.

Current applications

Burn and wound dressing , water filtration devices, paints, cosmetics quoting lubricants, textiles, memory storage devices, Medical diagnosis , display , sensors , drug delivery, composite material, biomaterial, Nano- array , protect armor powerful computers, chem - bio suits, chem – bio sensors.

Future Applications

- 2011-15 – nanobiomaterials, microprocessors, new catalysts, portable energy cells, solar cells, tissue/organ regeneration, smart implants
- 2016 and beyond – molecular circuitry, quantum computing, new materials, fast chemical analyses

Microscopy

Scientists use microscopes to visualize cells which are too small to see by naked eyes .

In light microscope (LM) : visible light is passed through the Specimen and then through the glass lenses .

Lenses reflect /bent the light so the image is magnified .



There are other microscopes rely on using Electron beams to visualize a specimen instead of using light beams called Electron microscope.

Two types of Electron microscopes

1. **Scan Electron Microscope (SEM)** : use Electrons beams the focus on the surface on specimen, the electrons will reflect without penetrate the sample , by Analyzing the Scattering of Electrons 3 D image will form .
2. **Transmission electron microscope (TEM)** : focus a beam of electrons that go through the Specimen and penetrate it , mainly used to study the internal structure of the cell , viruses , and molecules at micro and Nano scale .

Agro- Nanotechnology

(Nano technology in Agriculture)

Nano materials is a bridge between a Bulk material and atoms or molecules.

Bulk materials: constant physical properties

Nano materials : Quantum Confinement effect

When the metal convert from bulk stat into Nano particles new unusual properties arise which are different when the same material is in bulk form .

For example : Gold (Au)

Bulk gold : melt at 1064 C°

Gold Nano particles : melt at 300 C°

Copper (Cu)

Bulk copper: malleable/ Ductile

Copper Nano particles: super hard

Silver (Ag)

Bulk silver : non toxic

Silver Nano particles: anti- viral

→ Nano materials have higher surface- volume ratio compare to Bulk materials this could be exploit in Catalysis and detection.

Improve Agriculture by Nanotechnology

1. Increase in Productivity

- Development of Nano fertilizers : for immobilization of nutrients and their release in soil , such systems have advantage to minimize leaching while improving nutrients up take by plants and reducing the nitrogen in irrigated water .
- Development of Nano pesticides: increase the solubility, enhancing the resistant against Hydrolysis.
- ❖ Calcium borate Nanoparticles (synthetic fertilizer)Using on lettuce plant show an increase 2.7 fold in shoot length and 1.9 fold in root length .
- ❖ Copper , Silver , Zinc nanoparticles (synthetic Anti-fungi)
Depend on the concentration, many concentrations were tested choose the best concentration , it might toxic for plant at certain concentration , it might show degree of effects at different conditions and the best effect at certain concentration ,it might be non affect at certain concentration .

2- Soil quality improvement

- Enhancement of the soil water Capacity
- Increase soil remediation Capacity

Nano zerovalent iron is the most widely used Nano technologies in soil remediation.

Nano Zerovalent iron immobilize cadmium, copper and Nickel which are a heavy metals have a Toxic effects on plant .

3- plant growth stimulation

- Enhancement of elements uptake .
- Enhancement of nutrients use .

The impact of Nano material on plant depend on their composition, concentration, size , surface charge and physical and chemical properties.

An experiment was conducted using Zinc oxide Nano particles on tomato , after 45 day Tomato Show increase growth and photosynthetic rate .

4- smart monitoring

- Nano sensors: help farmers in maintain farms with precise control and report timely needs of plants .

Nano sensors could demonstrate their potential in Manning all the phases food supply change from crop cultivation and harvesting to food processing, transportation , packaging and distribution.

Ok there are a lot of Advantages of using Nanoparticles, but let ask a Two Questions

1- What is the fate of these particles after enter the organisms ?

2- What is the impact of these particles on environment ?

